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FEDERAL SPECIFICATION

HAND CLEANER, DUPLICATING INK; HAND CLEANER, GREASE

This specification was approved by the Commissioner, Federal Supply Service, General Services Administration, for the use of all Federal agencies.

1. SCOPE AND CLASSIFICATION

1.1 Scope. This specification covers two types of hand cleaners, with two classes under each type.

1.2 Classification.

1.2.1 *Types and classes.* Hand cleaner in accordance with this specification shall be furnished in the following types and classes, as specified (see 6.2).

Type I—Hand cleaner, duplicating ink
Class 1—Emulsified lotion form
Class 2—Cream form

Type II—Hand cleaner, grease
Class 1—Emulsified lotion form
Class 2—Cream form

2. APPLICABLE SPECIFICATIONS, STANDARDS, AND OTHER PUBLICATIONS

2.1 Specifications and Standards. The following specifications and standards, of the issues in effect on date of invitation for bids, form a part of this specification to the extent specified herein:

Federal Specifications:
UU-P-228—Paper Set, Duplicating, Master (Gelatin-Process and Spirit-or-Liquid-Process).
UU-T-591—Towels, Paper.
PPP-B-636—Box, Fiberboard.
PPP-C-96—Cans, Metal, 28-Gage and Lighter.

Federal Standards:
Fed. Std. No. 102—Preservation, Packaging, and Packing Levels.
Fed. Std. No. 123—Marking for Domestic Shipment (Civilian Agencies).
Fed. Test Method Std. No. 791—Lubricants, Liquid Fuels, and Related Products; Methods of Testing.

(Activities outside the Federal Government may obtain copies of Federal Specifications, Standards, and Handbooks as outlined under General Information in the Index of Federal Specifications, Standards, and Handbooks and at the prices indicated in the Index. The Index, which includes cumulative monthly supplements as issued, is for sale on a subscription basis by the Superintendent of Documents, U. S. Government Printing Office, Washington 25, D. C.

(Single copies of this specification and other product specifications required by activities outside the Federal Government for bidding purposes are available without charge at the General Services Administration Regional Offices in Boston, New York, Washington, D. C., Atlanta, Chicago, Kansas City, Mo., Dallas, Denver, San Francisco, and Auburn, Wash.

(Federal Government activities may obtain copies of Federal Specifications, Standards, and Handbooks and the Index of Federal Specifications, Standards, and Handbooks from established distribution points in their agencies.)

Military Standards:
MIL-STD-105—Sampling Procedures and Tables for Inspection by Attributes.
MIL-STD-129—Marking for Shipment and Storage.

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(Copies of Military Specifications and Standards required by contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

2.2 **Other publications.** The following document forms a part of this specification to the extent specified herein. Unless otherwise indicated, the issue in effect on date of invitation for bids shall apply.

Federal Food, Drug, and Cosmetic Act and General Regulations for Its Enforcement.

(Copies of the Federal Food, Drug, and Cosmetic Act and General Regulations for Its Enforcement may be obtained upon application, accompanied by money order, coupon or cash to the Superintendent of Documents, Government Printing Office, Washington 25, D. C. Prices may be obtained from the Superintendent of Documents.

3. REQUIREMENTS

3.1 **Material.** The detergents, emulsifiers, solvent, and skin conditioning ingredients used in the manufacture of hand cleaner shall be commercial grade or superior. The finished product with or without antiseptic chemicals shall be suitable for use on the human skin.

3.2 **Stability.** Hand cleaner shall be smooth-textured, stable, homogeneous material, and shall remain smooth and show no separation, rancidity or loss of effectiveness when tested as specified in 4.5.1.

3.3 **pH value.** The pH value at 25° ± 5° C. of a 5.0 percent by weight, aqueous solution of the cleaner shall have a value of 4.5 to 10.3 when tested as specified in 4.5.2.

3.4 **Grit.** Cleaner shall be free of grit and abrasives when tested as specified in 4.5.3.

3.5 **Chlorinated solvents.** Cleaner shall be free of chlorinated solvents. When specified, the manufacturer shall certify that his product complies with this requirement (see 6.2).

3.6 **Specific gravity.** Cleaner shall have a specific gravity ranging from 0.85 to 1.15 when tested as specified in 4.5.4.

3.7 **Residue on the skin.** Cleaner shall not leave a sticky or undesirable coating on the skin when tested as specified in 4.5.6.2 and 4.5.6.3.

3.8 **Odor.** Cleaner shall be free of odors characteristic of kerosene and similar petroleum products. Cleaner shall be free from objectionable odor and shall not leave an objectionable odor on the skin when tested as specified in 4.5.5.

3.9 **Performance.** Type I shall effectively remove duplicating ink from hands and type II shall effectively remove greasy soil from hands, when tested as specified in 4.5.6.

3.10 **Flash point.** The flash point of the cleaner shall be not lower than 40° C. (104° F.) when tested as specified in 4.5.7.

3.11 **Effect on the skin.** Cleaner shall not irritate nor sensitize the skin when tested as specified in 4.5.8.

3.12 **Container size.** Hand cleaner shall be furnished in half-pint or pint size cans, as specified (see 6.2).

3.13 **Federal Food, Drug, and Cosmetic Act.** Cleaner shall comply with the provisions of Chapter VI of the Federal Food, Drug and Cosmetic Act, as applicable (see 6.3).

3.14 **Workmanship.** Cleaner furnished in accordance with this specification shall be of the finest quality in every respect and shall possess no deficiency that will affect serviceability.

4. SAMPLING, INSPECTION, AND TEST PROCEDURES

4.1 The supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified, the supplier may utilize his own or any other inspection facilities and services acceptable to the Government. Inspection records of the examination and tests shall be kept complete and available to the Government as specified in the contract or order. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are

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deemed necessary to assure supplies and services conform to prescribed requirements.

4.2 **Lot.** A lot shall consist of all hand cleaner of same type, class, and container size, manufactured as one batch and offered for delivery at one time. Unless otherwise specified, the unit of product shall be one container of hand cleaner.

4.3 **Sampling.**

4.3.1 *Sampling of shipping lot.* The assembled lot of packed and marked containers of hand cleaner shall be examined for compliance with the applicable requirements of section 5 of this specification. The unit of product shall be one sealed and marked shipping container. The number of shipping containers selected for examination from each lot offered for delivery shall be in accordance with table III–B inspection level L–4 of MIL–STD–105.

4.3.2 *Sampling for examination.* Sample containers of hand cleaner shall be selected at random from each lot in accordance with table III–B inspection level L–8 of MIL–STD–105.

4.3.3 *Sampling for test.* A random sample of hand cleaner containers shall be selected from each lot in accordance with table III–B, inspection level L–4 of MIL–STD–105.

4.4 **Examination.** Shipping containers selected in accordance with 4.3.1 and containers of hand cleaner selected in accordance with 4.3.2 shall be examined for the applicable defects listed in table I and any other requirements of this specification that can be determined by visual examination. The acceptable quality level (AQL) for shipping containers shall be 4.0 percent defective. The AQL for containers of hand cleaner shall be 2.5 percent defective.

4.4.1 *Lot acceptance tests.* Samples of hand cleaner selected in accordance with 4.3.3 shall be subjected to the tests specified herein. Any sample failing the tests specified herein is sufficient cause for rejecting the lot.

4.5 **Test procedures.**

TABLE I.—*Classification of defects*

Examination	Defect
	Shipping Containers
Packing	Not type or class specified.
	Closure not secure.
	Crushed, torn or punctured container (affecting serviceability).
Marking	Illegible or incorrect.
Packaging ..	Not as specified.
	Package count less than specified.
	Package not snug fitting.
	Labels do not bear required data.
	Hand Cleaner
Form	Not as specified

4.5.1 *Heat stability.*

4.5.1.1 *High temperature.* Place an unopened issue package containing samples in an oven maintained at 50° ± 2° C. After 16 hours remove from oven and allow sample to cool to room temperature. After 48 hours open container and examine contents for a separation of oil, water, or solid matter; note odor, which should not be objectionable. Use a portion of product as specified in 4.5.6 to determine its effectiveness in the removal of soil from the skin.

4.5.1.2 *Low temperature.* Place an unopened issue package of cleaner in a cold chamber at 0° ± 2° C. After 16 hours, remove from cold chamber and allow to come to room temperature. Eight hours later, open container and examine contents for separation of oil, water, or solid matter; note odor, which should not be objectionable.

4.5.2 *pH value.* Determine the pH, at 25° ± 1° C. of a 5 percent (by weight) solution in distilled water, using a pH meter with a glass electrode. Calibrate the pH meter before use by checking against a standard buffer solution of pH 7.0.

4.5.3 *Grit.* Spread approximately 1 gram of cleaner on a clean microscope slide. Place a second slide over film of cleaner. Press the two slides together with a rotary motion. There shall be no evidence of grit. After wiping free of cleaner, glass shall show no scratch marks.

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4.5.4 *Specific gravity.* Determine specific gravity in a calibrated centrifuge tube. Use the centrifuge to pack the tube. The centrifuge shall be operated at a speed of approximately 1725 r.p.m. for a period of three minutes using a 15 milliliter tapered centrifuge tube and at a temperature of 20° ± 2° C.

4.5.5 *Odor.* Open a fresh container and note the odor of the contents. Use 2 grams of the cleaner on the hands following the direction for use given on the container. Note residual odor on hand after 5 minutes have elapsed and after 60 minutes.

4.5.6 *Performance.*

4.5.6.1 *"Exposed" test.* Fill a shallow dish about 8 centimeters (cm) in diameter by 1 cm. in depth (such as a Petri culture dish) with the sample. Allow to stand at 31° ± 1° C. and at a relative humidity of 50 plus or minus 5 percent for 5 hours. The "exposed" cleaner shall be in condition suitable for use. Retain the "exposed" cleaner for the applicable performance test specified hereinafter.

4.5.6.2 *Type I (to be tested on hands of female subjects).* First prepare the hands by washing them thoroughly, using soap and water, followed by thoroughly drying with a towel. Without soiling the hands, wait for fifteen minutes before proceeding with the application of the testing material. Cut a two-inch duplicating master set conforming to type II, grade B of UU–P–228. Place this two-inch square of paper dye-face down toward palm of left hand, and with the fingers of the right hand rub the paper back and forth and in rotary motion over the palm of the left hand for a period of 30 seconds. Rub as much of the dye off of the paper onto the palm of the hand as practicable in this period of time. Rub in all directions. Take a clean sheet of paper toweling conforming to UU–T–591, and rub the palm of the left hand such that the excess ink is wiped off and the dye which remains is rubbed well into the hand. Rub in this matter for another 30 seconds. After fifteen minutes (at room temperature of 23° ± 2° C. (73° ± 3.5° F.) and a relative humidity of 50 plus or minus 2 percent) apply 1.0 plus or minus 0.1 gram of cleaner to the soiled left palm with finger of clean hand; rub cleaner over stained area with finger of the clean hand for exactly one minute. Wipe palm of stained hand with a paper towel conforming to UU–T–591, removing all of the excess cleaner from the left hand. Reapply 1.0 plus or minus 0.1 gram of fresh cleaner rubbing the palm of the left hand for exactly one minute as above. Wipe the hand clean of cleaner with a fresh paper towel as described above. Evidence of stain remaining on palm indicates failure to meet specification and shall be cause for rejection. Note presence of undesirable residue on skin or unpleasant "feel" of skin on which cleaner has been used. (Tests shall be run separately on both fresh and "exposed" samples of cleaner).

4.5.6.3 *Type II (to be tested on hand of male subjects).*

4.5.6.3.1 *Standard test soil.* The standard test soil shall have the following composition:

Standard Test Soil:

1 quart SAE #20 lubricating oil
½ pound lime grease
4 ounces carbon, lamp black
6 ounces charcoal black

Mix the above ingredients thoroughly and use for test as described in 4.5.6.3.2.

4.5.6.3.2 *Procedure.* Before proceeding with the test, the hands should be washed thoroughly, using soap and water, then dried thoroughly with a towel, and without soiling the hands, wait for a period of fifteen minutes before proceeding with the test. Rub 0.1 gram of the above "standard test soil" into the skin of the back of one hand only, being sure to rub well into the knuckles, creases of the skin, fingernails, and cuticles. Using a paper towel conforming to UU–T–591, rub the test soil with the towel so as to remove the excess and to rub the soil which remains well into the skin, knuckles, creases of the skin, fingernails, and cuticles. Apply 1.0 plus or minus 0.1 gram of cleaner to soiled hand, rubbing with other hand for exactly one minute, using lengthwise, crosswise, and rotary

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strokes. Using a fresh paper towel as described above, wipe the excess cleaner from the back of the hand. Repeat the application of 1.0 plus or minus 0.1 gram of cleaner to the back of the soiled hand repeating the rubbing for exactly one minute as described above; wipe the back of the hand free of all cleaner and soil which can be removed with the towel. There shall be no appreciable difference in color or shade between back of hand which has been soiled, then treated with cleaner, and back of hand which has not been soiled. Note presence of undesirable residue on skin or unpleasant "feel" of skin on which cleaner has been used. Failure of cleaner to remove all visible evidence of soil shall be cause for rejection. (Test shall be run on both fresh and "exposed" samples of cleaner, separately.)

4.5.7 *Flash point.* Flash point shall be determined in accordance with method 1102.8 of Fed. Test Method Std. No. 791.

4.5.8 *Dermatitis test.* Preliminary animal tests to be conducted to detect the presence of primary irritating and sensitizing properties. Clip the fur of 5 adult albino rats by means of small animal clippers to a length less than 1/16 inch over an area of approximately 2 square inches in the region between and just below the shoulders. The animals should then be segregated in individual cages to eliminate the possible injury by contact with the claws of other animals. Only those showing absence of lesions or excoriations in the bare area at the end of 2 days should be utilized. Apply approximately ¼ cc. of compound to the bare area of the rats, and examine for visible irritation daily for 1 week. If there is no irritation to the skin at the end of one week, repeat the above test after a 10 to 14 day waiting period. If no irritation is visible, it shall be considered that the compound will not irritate or sensitize the skin.

5. PREPARATION FOR DELIVERY

For civil agency procurement, the definitions and applications of the levels of packaging and packing shall be in accordance with Fed. Std. No. 102.

5.1 Packaging. Packaging shall be level A or C (see 6.2).

5.1.1 *Level A.* Type I or type II hand cleaner shall be furnished in half-pint or pint size cans, as specified, conforming to type VI of PPP–C–96. Cans shall be furnished with slip or threaded covers, as specified. The threaded cover shall have a minimum of one and one-half threads. The top portion of the periphery of the cover shall be knurled to permit easy removal.

5.1.2 *Level C.* Packaging shall be in accordance with the manufacturer's commercial practice.

5.2 Packing. Packing shall be level A, B, or C (see 6.2).

5.2.1 *Level A.* Seventy-two half-pint cans of hand cleaner of like type or forty-eight pint cans of hand cleaner of like type shall be packed in containers conforming to PPP–B–636, class 2. Half-slotted, snug-fitting separators or partitions of full height shall be used. A full pad shall be placed between the tiers. Partitions and pad shall be of the same material as the container. Closure shall be in accordance with the appendix of this specification.

5.2.2 *Level B.* Seventy-two half-pint cans of hand cleaner of like type or forty-eight pint cans of hand cleaner of like type shall be packed into containers conforming to PPP–B–636, class 1. Half-slotted, snug-fitting separators or partitions of full height shall be used. A full pad shall be placed between the tiers. Partitions and pad shall be of the same material as the container.

5.2.3 *Level C.* The hand cleaner, packaged as described in 5.1 shall be packed in a manner to insure carrier acceptance and safe delivery at destination. The containers shall be in accordance with the rules or regulations of carriers applicable to the mode of transportation.

5.3 Marking.

5.3.1 *Military agencies.* In addition to any other marking required by the contract, the containers shall be marked in accordance

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with the requirements of MIL–STD–129, as applicable.

5.3.2 *Civil agencies.* Containers shall be marked in accordance with the requirements of Fed. Std. No. 123.

6. NOTES

6.1 Intended use. Type I hand cleaner is intended for removing duplicating inks. Type II hand cleaner is intended for removing grease, grime and stains.

6.2 Ordering data. Procurement documents should specify the following:

(*a*) Title, number and date of this specification.
(*b*) Type and class (see 1.2.1).
(*c*) Manufacturer's certification, when required (see 3.5).
(*d*) Size of container (see 3.12); type of closure (see 5.1.1).
(*e*) Applicable level of packaging and packing (see 5.1 and 5.2).

6.3 Matters pertaining to compliance of products under the Federal Food, Drug and Cosmetic Act should be referred to the Department of Health, Education and Welfare, Food and Drug Administration, Washington 25, D. C.

6.4 Items procured under this specification for military use are to be limited to the variety shown on the applicable military standards. Personnel of the military departments are requested to refer to these documents for guidance.

6.5 Transportation description. Transportation description and minimum weights applicable to this commodity are:

Types I and II, Class 1
Rail:
Soap not otherwise indexed by name, liquid.
Carload minimum weight 36,000 pounds.
Motor:
Soap, not otherwise indexed, liquid.
Truckload minimum weight 36,000 pounds, subject to Rule 34, National Motor Freight Classification.

Types I and II, Class 2
Rail:
Soap, not otherwise indexed, other than liquid.
Carload minimum weight 36,000 pounds.
Motor:
Soap not otherwise indexed, other than liquid.
Truckload minimum weight 36,000 pounds, subject to Rule 34, National Motor Freight Classification.

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